

HEMARUKA AGCM, AB, Canada

WMO: 717930

Lat: 51.7797N

Lon: 111.2119W

Elev: 780

StdP: 92.30

Time Zone: -7.00 (NAM)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-32.3	-29.1	-35.8	0.1	-32.2	-32.4	0.2	-29.1	11.4	-15.7	9.8	-17.5	2.0	190	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.0	30.1	16.3	27.9	16.3	25.9	15.5	18.5	25.0	17.5	24.3	16.6	23.6	5.1	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.3	12.8	21.3	15.1	11.8	20.2	14.1	11.0	19.0	55.7	24.9	52.3	24.5	49.6	23.5	22.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
12.1	10.4	8.9	-36.2	33.2	2.3	2.8	-37.8	35.2	-39.1	36.8	-40.4	38.4	-42.1	40.4		
			-36.3	20.8	2.3	1.2	-37.9	21.6	-39.2	22.3	-40.5	22.9	-42.1	23.8		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.7	-11.0	-12.4	-4.9	3.0	10.3	14.4	17.3	16.5	11.4	3.9	-4.8	-12.1	(6)
	DBStd	12.43	8.07	7.96	7.99	5.33	4.25	3.26	2.95	3.37	4.63	5.13	7.22	8.26	(7)
	HDD10.0	3375	650	627	462	216	50	4	0	1	37	199	444	684	(8)
	HDD18.3	5753	908	861	720	461	251	124	55	76	212	449	694	943	(9)
	CDD10.0	718	0	0	0	5	58	136	227	203	79	9	1	0	(10)
	CDD18.3	53	0	0	0	0	1	6	23	20	4	0	0	0	(11)
	CDH23.3	1138	0	0	0	3	53	124	417	389	148	4	0	0	(12)
	CDH26.7	321	0	0	0	0	6	26	125	117	47	0	0	0	(13)
(14) Wind	WSAvg	4.2	3.9	3.7	4.4	4.7	4.6	4.5	4.3	4.1	4.0	4.8	3.9	3.9	(14)
(15) Precipitation	PrecAvg	355	16	10	15	22	40	73	59	48	28	14	14	12	(15)
	PrecMax	465	42	23	36	51	83	153	148	87	74	34	53	19	(16)
	PrecMin	264	3	1	4	3	4	14	23	11	3	5	1	2	(17)
	PrecStd	66	9	5	8	13	24	32	27	25	18	7	12	5	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.3	5.8	14.9	22.0	27.2	30.0	32.5	32.7	30.9	22.8	14.9	6.1	(19)
		MCWB	2.9	2.7	7.8	9.6	13.6	16.6	17.7	16.4	15.0	12.1	7.2	3.2	(20)
	2%	DB	4.0	3.0	11.3	18.9	24.8	26.5	30.0	29.6	27.6	18.8	10.2	3.7	(21)
		MCWB	1.8	0.9	5.3	9.1	12.5	15.4	17.2	16.3	14.6	10.4	5.1	1.4	(22)
	5%	DB	2.3	1.1	8.1	15.6	22.7	24.4	27.6	27.4	24.3	15.6	6.8	1.4	(23)
		MCWB	0.5	-0.4	3.6	6.8	11.6	14.7	17.0	16.1	13.6	8.7	3.3	-0.7	(24)
	10%	DB	0.3	-0.9	4.8	12.6	20.1	22.4	25.4	25.3	21.2	12.6	4.2	-1.4	(25)
		MCWB	-1.0	-2.0	1.8	5.5	10.5	14.0	16.3	15.6	12.5	7.1	1.6	-2.7	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.2	2.9	8.1	10.9	14.6	18.4	20.9	19.6	16.6	13.3	8.2	3.5	(27)
		MCDB	5.9	5.7	14.6	20.5	24.8	24.9	29.2	25.8	26.8	21.9	14.5	5.5	(28)
	2%	WB	1.8	1.0	5.7	9.0	13.3	16.9	19.0	18.2	15.2	11.1	5.6	1.4	(29)
		MCDB	3.7	2.7	10.4	18.3	22.6	24.1	25.1	24.9	24.9	16.4	9.8	3.6	(30)
	5%	WB	0.7	-0.3	3.9	7.3	12.4	15.8	17.9	17.2	14.0	9.2	3.7	-0.5	(31)
		MCDB	2.3	1.1	7.8	14.4	21.0	22.3	24.4	24.5	22.7	14.9	6.5	1.3	(32)
	10%	WB	-1.0	-2.0	2.0	6.0	11.2	14.8	17.0	16.2	12.9	7.5	1.9	-2.7	(33)
		MCDB	0.3	-0.9	4.5	12.1	18.9	20.7	23.6	23.6	20.9	12.3	3.9	-1.4	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	10.5	10.9	10.9	12.7	14.1	12.9	14.0	14.3	14.6	12.9	10.5	10.0	(35)
		MCDBR	10.9	12.0	14.4	18.4	18.3	16.3	18.0	18.6	20.5	17.7	13.7	12.4	(36)
		MCWBR	9.0	10.1	8.9	9.3	8.3	7.2	7.4	7.4	9.0	9.8	9.2	10.0	(37)
	5% WB	MCDBR	10.9	11.7	13.9	16.8	16.1	13.9	14.7	15.4	18.8	16.7	13.0	12.0	(38)
		MCWBR	9.2	10.0	8.9	9.0	7.6	6.8	6.9	7.1	8.7	9.4	8.9	9.8	(39)
(40) Clear-Sky Solar Irradiance	taub		0.268	0.321	0.289	0.343	0.367	0.375	0.362	0.380	0.324	0.285	0.259	0.258	(40)
	taud		2.310	2.163	2.453	2.318	2.297	2.324	2.397	2.315	2.451	2.536	2.477	2.306	(41)
	Ebn at Noon		749	779	919	894	885	879	885	846	865	845	777	696	(42)
	Edh at Noon		62	97	88	114	123	121	110	114	88	66	52	53	(43)
(44) All-Sky Solar Radiation	RadAvg		1.15	2.21	3.60	4.83	5.83	6.04	6.45	5.25	3.75	2.27	1.19	0.87	(44)
	RadStd		0.11	0.19	0.27	0.41	0.39	0.38	0.28	0.40	0.35	0.29	0.10	0.08	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (64 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page